

Day-Ahead Imbalance Price Forecasting: What Was Tested and What It Shows

Recurrent encoders and market-rule blocks in a 96-horizon MRINN extension

30 AUGUST 2026 TEST WINDOW 2025 9 MODELS · 4 ROLLING FOLDS 34,521 ORIGINS 7.7 H CPU

Nine forecasting models were trained and evaluated under one identical protocol to answer two questions the project had left open: **does a recurrent encoder beat MRINN's Dense encoder at a day-ahead horizon**, and **do the differentiable market-rule blocks contribute anything measurable at all**. Four defects in the prior evaluation were found and corrected first, without which no comparison would have been valid. The recurrent-encoder claim is confirmed and now statistically significant; the market-rule claim survives only in combination with the GRU encoder, and is null elsewhere. Section 5 assesses five supervisor suggestions against the same data, and records which hold, which need correction, and which are blocked.

1 What was tested

TASK AND DATA	PROTOCOL
Austrian/German balancing-market imbalance price at 15-minute resolution (<code>imbalance_data.csv</code> , 140,249 rows, 2022–2026). At each origin the model predicts all 96 horizons — 15 minutes to 24 hours — in a single forward pass. 17 input signals; no lagged price among them.	Rolling 12-month training windows retrained quarterly, four folds covering all of 2025. Per-fold scalars, inverted per fold before pooling. $T = 32$ (8 h history), stride 2 on training rows only; batch 256, constant LR with plateau decay, early stopping on validation loss. Single seed (42).

The nine models

The design is a **2×3 grid plus references**. Every model shares the data pipeline, horizon embedding, known-future covariate path, day-ahead outage dropout and — except LQR and XGBoost — the hierarchical quantile head. Reading *down* a column isolates the encoder; reading *across* a row isolates the market-rule blocks.

ENCODER	WITH MARKET RULES		RULES REMOVED	ISOLATES
Dense	MRINN (6,135 p)		MLP (7,533 p)	Row = value of the rule blocks Column = value of the encoder
LSTM	MR-LSTM (5,895 p)	LSTM (7,261 p)		
GRU	MR-GRU (5,055 p)	GRU (6,309 p)		
References	AttnBiLSTM (15,302 p) · XGBoost (96 boosters/fold) · LQR (linear, ridge) · climatology · seasonal naive			

Parameter counts are exact. XGBoost has no comparable weight count. LQR and XGBoost deliberately omit the hierarchical quantile head, so that quantile crossing can be observed rather than assumed away.

Metrics

AQL (average pinball loss over $q = 0.10, 0.25, 0.50, 0.75, 0.90$) is the headline; **MAE** and **RMSE** of the median; **AQCR** (quantile-crossing rate); 80% interval coverage. Reported at $h = 1$ (15 min), $h = 4$ (1 hour) and $h = 96$ (1 day), and averaged over all 96.

2 Four corrections made before any comparison

Each of these invalidated or distorted the previous results. They are reported because they change published numbers, not as housekeeping.

DEFECT	MEASURED EXTENT	CONSEQUENCE
Day-ahead look-ahead leak	14.95% of day-ahead cells; 54.2% of origins	Models consumed prices not yet auctioned. Gate closure now enforced.
Models never converged	Every prior run stopped at the epoch ceiling (119–120 of 120)	Batch 1024 gave 18 updates/epoch — 64% of requirement. Now 256.

DEFECT	MEASURED EXTENT	CONSEQUENCE
Baseline weaker than a constant	Seasonal-naive RMSE 547 $\approx \sqrt{2} \times$ price SD 392	"-32% vs naive" overstated. Climatology added as the real reference.
RMSE averaged across folds	321 averaged vs 390 pooled	RMSE is a root of a mean of squares; understated by ~17%.

The leak is structured, not diffuse: zero at horizons below $h = 45$, rising to 54.2% of origins at $h = 96$, and confined entirely to origins before 13:00 local — the signature of a 13:00 auction publication gate.

ON THE LEAK'S SEVERITY
The leaked price is genuinely informative (a legitimate same-clock-time substitute explains only $R^2 = 0.26$ of it), but it barely predicts the target: univariate $R^2 = 0.045$ against 0.011 for the substitute. Closing it was necessary for validity and did not overturn the ranking.

3 Results

MODEL	RULES	AQL	MAE	RMSE	AQCR	PARAMS	VS CLIM.
MR-GRU	yes	25.574	65.01	389.4	0.00	5,055	-12.3%
MR-LSTM	yes	25.670	65.28	389.4	0.00	5,895	-11.9%
LSTM	no	25.746	66.20	389.6	0.00	7,261	-11.7%
MRINN	yes	25.819	66.22	390.0	0.00	6,135	-11.4%
AttnBiLSTM	no	25.829	66.52	389.6	0.00	15,302	-11.4%
GRU	no	25.860	66.79	390.0	0.00	6,309	-11.3%
MLP	no	25.871	66.17	389.6	0.00	7,533	-11.2%
XGBoost	no	26.346	67.44	389.9	2.75	—	-9.6%
LQR	no	28.022	74.96	391.8	15.75	538,080	-3.9%
Climatology	—	29.150	75.20	393.8	0.00	—	0.0%
Seasonal naive	—	39.870	96.03	547.0	0.00	—	+36.8%

Pooled over all four folds — scored once on concatenated per-fold predictions, not averaged. No model emitted a single prediction beyond $\pm 5,000$ €/MWh. Every model converged via early stopping; none stopped at its epoch ceiling.

3.1 The three reporting horizons

HORIZON	BEST MODEL	AQL	MAE	RUNNER-UP	THIRD
15 min ($h = 1$)	XGBoost	23.010	56.44	GRU 24.609	LSTM 24.894
1 hour ($h = 4$)	MR-GRU	25.220	63.35	GRU 25.231	LSTM 25.236
1 day ($h = 96$)	MR-GRU	25.957	66.41	LSTM 26.002	GRU 26.048

All read off the same forward pass, so the three rows are mutually consistent and required no per-horizon retraining.

3.2 Do the market-rule blocks earn their place?

ENCODER	WITH RULES	WITHOUT	Δ AQL	$\Delta\%$	FOLDS WON	P (DAILY)
Dense	MRINN	MLP	-0.051	-0.20%	2/4	0.255 — n.s.
LSTM	MR-LSTM	LSTM	-0.076	-0.29%	2/4	0.961 — n.s.
GRU	MR-GRU	GRU	-0.286	-1.10%	3/4	0.0002

Paired by fold. Fold difficulty spans AQL 17 to 42 — several times any model effect — so pooled means would be dominated by which quarters a model was scored on.

3.3 Significance

Diebold–Mariano was run two ways: Newey–West at lag 96, and on non-overlapping daily origins. Adjacent origins share 95 of their 96 targets, so the standard HAC correction overstates significance; where the two disagree, the daily test is believed.

COMPARISON	Δ AQL	P (NEWY-WEST)	P (DAILY)	VERDICT
MR-GRU – MRINN	–0.245	0.0002	0.0017	Significant
MR-GRU – MR-LSTM	–0.096	0.0002	0.0014	Significant
MR-GRU – AttnBiLSTM	–0.255	<0.0001	0.0050	Significant
MR-GRU – XGBoost	–0.772	<0.0001	<0.0001	Significant
MR-LSTM – MRINN	–0.148	0.0285	0.3689	Overlap artefact

The final row is the reason both tests are reported: significant under Newey–West, not significant once overlapping origins are removed.

4 Conclusions

- 1

The recurrent-encoder claim holds, and is now significant.

MR-GRU beats MRINN by 0.245 AQL ($p = 0.0017$ on non-overlapping origins), wins all four folds, and is the best model at both 1 hour and 1 day — on **5,055 parameters, fewer than any other neural model tested**. This overturns the project's own earlier conclusion of a tie at day-ahead, which was recorded before the leak and convergence defects were found.
- 2

The market-rule blocks are worth far less than assumed.

They are statistically null at the Dense and LSTM encoders and significant only with the GRU. Averaged across encoders they buy 0.53% of AQL. Their defensible value is **parameter efficiency**, not accuracy: MR-GRU matches or beats every rules-free model on 20% fewer parameters.
- 3

GRU and LSTM are not interchangeable here.

MR-GRU beats MR-LSTM significantly on both tests ($p = 0.0014$), so the cell type is a real choice rather than a matter of taste — and the cheaper cell is the better one.
- 4

Gradient boosting wins the short horizon outright.

XGBoost is best of all nine at 15 minutes (AQL 23.01 against MR-GRU's 24.96, a 7.8% margin) and worst at 24 hours. It exploits recent lags but has no horizon structure, and needs 96 separate boosters. Any claim of a single model for the whole day must concede this point.
- 5

Zero quantile crossing is a property of the head, not of the encoder.

Every model carrying the hierarchical quantile head scores AQCR = 0. The two models without it do not: XGBoost 2.75, LQR 18.07. This was previously inherited from the source paper as an assumption; it is now demonstrated.
- 6

The honest headline is smaller than the one previously reported.

Against climatology on matched rows, the best model is **–12.3% AQL and –13.4% MAE** — not –32% against seasonal naive, a baseline that itself loses to predicting a constant. RMSE sits at parity with a constant predictor for every model: these models forecast the calm bulk of the distribution and capture essentially none of the spike variance.

5 Assessment of the supervisor's suggestions

Five suggestions were raised. Each was checked against the data rather than argued from principle; verdicts and the evidence behind them follow.

REF	SUGGESTION	VERDICT
S1	Add an imbalance-netting block modifying P_RE upstream	Adopt
S2	Block diagram drawn as RE → EX → SC → price	Correct before publication
S3	Report sMAPE alongside MAE and RMSE	Push back
S4	120 epochs was not converging — try 200	Concede
S5	“Need to verify the signs” on the INC series	Blocking — needs his answer

S1 — Netting into P_RE is the right block, but its influence is shrinking

Modifying P_RE is correct: it is decisive on the majority of intervals while P_SC is nearly inert. Placing netting *upstream* of the rule rather than adding a fourth parallel component is also correct — netting changes the volume requiring activation, it does not itself set a price.

YEAR	INTERVALS	P_RE	P_EX	P_SC
2022	35,040	82.4%	17.5%	0.1%
2023	35,040	71.6%	27.9%	0.4%
2024	35,136	67.7%	31.6%	0.7%
2025 — the test window	35,033	53.8%	44.7%	1.4%

Share of intervals where each rule component equals the settled price. P_RE's dominance has fallen every year while P_EX has risen to meet it.

Observation. Over the whole record P_RE decides 68.9% of intervals, but on the year actually tested it decides **53.8%**. A netting improvement will therefore show on roughly half the test intervals, not two-thirds, and the write-up should quote the 2025 figure rather than the flattering whole-record one. Two checks before committing: `system_imbalance` is already **83.1%** explained by net activation alone ($r = +0.911$), so INC may add little; and inside the model these signals are 8-dimensional learned encodings, so the block is *netting-structured*, not literal netting, and should not be over-claimed as physical fidelity.

S2 — The block diagram contradicts the code

The diagram shows a series chain. The implementation computes the three components **in parallel** and blends the smooth minimum and maximum through a learned gate; the only real dependency is $P_EX \rightarrow P_SC$, because scarcity consumes `P_base` from the market-reference block.

Conclusion. Redraw before publication — a reviewer who opens the code will catch it. This is not cosmetic: the series diagram implies that netting entered at RE would propagate through EX and SC. It will not. Netting changes one of three parallel candidates and must then win the min/max gate to move the price at all, which is precisely why the declining P_RE share above matters.

S3 — Why sMAPE does not survive this price series

sMAPE divides by $(|actual| + |forecast|) / 2$, a denominator that is only well-behaved when prices are comfortably positive. Imbalance prices are not.

CONDITION	ROWS	SHARE	ACTUAL	FORECAST	ERROR	SMAPE
Price negative	2,372	6.77%	200	205	5	2.5%
Price exactly zero	640	1.83%	20	25	5	22.2%
$ price \leq 10 \text{ €/MWh}$	2,717	7.76%	2	7	5	111.1%
Any of the above	4,363	12.45%	-3	2	5	200.0%

2025 window, 35,033 intervals. Zeros make sMAPE undefined; the rest collapse its denominator. The union is exact, not the sum — the sets overlap.

One fixed €5 error, scored at four price levels. Identical forecast quality scores 2.5% to 200% depending only on where the price sits.

Conclusion. The headline would track the price level rather than forecast skill, driven by the 12.45% of intervals near or below zero. There is also a purpose mismatch: sMAPE exists to compare forecasts *across series of different scales*, and there is one series here, so scale-invariance buys nothing while costing interpretability. **Propose MASE instead** — scaled by in-sample seasonal-naïve error, it handles zeros and negatives and is standard in the forecasting literature. If sMAPE is still required, report it restricted to $|actual| > 10$ alongside the unrestricted figure; the gap between the two is itself the argument.

S4 — The convergence complaint was correct

He was right, and the evidence was already in the project's own output: with a ceiling of 120 epochs and patience 12, every run stopped at 119 or 120. Patience never fired, so training ended because the loop ran out rather than because the model stopped improving.

CONFIGURATION	UPDATES / EPOCH	EPOCHS	TOTAL UPDATES	OF REQUIREMENT
Previous — batch 1024	18	120	2,160	64%
His suggestion — batch 1024	18	200	3,600	106% — marginal
Adopted — batch 256	69	~49	3,381	comfortable

17,480 training rows per rolling fold at stride 2.

Conclusion. The diagnosis was right but the lever was batch size, not the epoch number: what a model learns from is gradient updates, and epochs only buy updates at a rate the batch size sets. His 200 would have sufficed but only just — the deepest fold needed 109 epochs plus confirmation. Every model in this report now stops via early stopping. Stated honestly, the old runs were not badly *wrong* — validation loss improved only 0.4% from epoch 20 to 49 — they were **unconverged and therefore indefensible**, which is a more precise complaint and a more defensible fix.

S5 — What “INC” refers to is genuinely ambiguous

His own note that the signs need verifying is well placed. In a German control-zone context INC plausibly denotes either the **IGCC imbalance-netting volume** — MW of opposing imbalance cancelled against neighbouring TSOs before activation — or the **reBAP incentive component** (*Anreizkomponente*), which is a price adder, not a volume. Different quantities, different units, different sign conventions.

Conclusion. Building against the wrong one yields a model that trains, converges and means nothing. This is the only item that cannot be resolved without him, and it is compounded by an unresolved contradiction in the project's own records: the dataset is described as the **Austrian** balancing market in the project log and as **German control-zone** in the audit. The bidding zone should be confirmed at the same time as the sign convention.

6 Limitations

STATED PLAINLY	NOT YET RESOLVED
• Single seed. Prior work measured seed noise at ± 0.11–0.26 AQL, which exceeds several differences reported here. The DM tests are across origins, not seeds, so they do not cover seed variation. • No spike skill. RMSE parity with a constant is a real limitation, not a presentational one.	• The dataset is described inconsistently as Austrian and as German control-zone. This does not affect the results above, but it must be settled before the proposed imbalance-netting extension, which depends on which market's INC series is meant. • Rule-component dominance is shifting: P_RE fell from 82.4% of intervals in 2022 to 53.8% in 2025 while P_EX rose to 44.7%.

7 Recommended next steps

1. **Repeat at two further seeds** for the four models that carry a claim (MR-GRU, MR-LSTM, MRINN, GRU). This is the single highest-value remaining run and the only way to convert "significant across origins" into "robust".
2. **Report the rules as an efficiency result**, not an accuracy one, and state the encoder-dependence explicitly rather than averaging it away.
3. **Settle the market and the INC definition** before building the netting block, and quote the 2025 P_RE share (53.8%) rather than the whole-record 68.9%.
4. **Replace sMAPE with MASE** (\$5, S3), and settle the bidding zone and INC definition (\$5, S5) before any netting work begins.

Reproduce: `benchmark_models.py` (training), `summarise_benchmark.py` (tables), `dm_test.py` (significance), `baselines.py` (climatology & naive), `audit_gate_closure.py` (leak measurement). Outputs in `results/`. All figures recomputed from `imbalance_data.csv`; nothing carried from earlier runs.